



INTEGRAL CALCULUS — FAST MCQ TRICKS

Class 12 Applied Maths — By Thunderstudy

Think of integration like reverse differentiation. If differentiation is breaking, integration is joining back. MCQs mostly test recognition!

⚡ 1. Recognition Trick (Most Powerful)

Instead of integrating fully, ask: **“Which function gives this when differentiated?”**

Example: $\int 2x \, dx$. Think: derivative of $x^2 = 2x$. Answer instantly $\rightarrow x^2 + C$.

👉 **MCQ Secret: Options already contain the answer — just match derivative.**

⚡ 2. Formula Spotting Trick

Integral	Direct Answer
$\int x^n \, dx$	$x^{n+1}/(n+1)$
$\int 1/x \, dx$	$\ln x $
$\int e^x \, dx$	e^x
$\int \sin x \, dx$	$-\cos x$
$\int \cos x \, dx$	$\sin x$

⚡ 3. “Derivative Inside” Trick (Chain Rule Reverse)

Look for pattern:

$$\int f'(x) \cdot f(x) \, dx = (f(x))^2/2$$

Example: $\int x(x^2+1) \, dx$. Derivative of (x^2+1) is $2x$. Answer: $(x^2+1)^2/2$ (approx). No expansion needed!

⚡ 4. Option Checking Method (MCQ GOLD 🏆)

Instead of solving integral: **Differentiate options.**

1. Pick an option.
2. Differentiate quickly.
3. If result matches question → DONE!

💥 **Works especially when the integral looks scary.**

⚡ 6. Definite Integral Mirror Trick

If limits are symmetric:

$$\int_{-a}^a f(x) \, dx$$

- Odd function ($x^3, \sin x$) → Result = 0 ✓
- Even function ($x^2, \cos x$) → Result = $2 \int_0^a f(x) \, dx$

🔥 **Many MCQs solved in 5 seconds.**

⚡ 7. Zero Trick

$$\int_a^a f(x) \, dx = 0$$

Same limits → Answer always 0. No thinking required!

⚡ 9. Log Substitution Pattern

If you see:

$$\int \left[\frac{f'(x)}{f(x)} \right] dx$$

Answer directly: $\ln |f(x)|$

5-Second MCQ Strategy

- **1** Look for formula match
- **2** Check derivative pattern
- **3** Observe limits symmetry
- **4** Differentiate options if confused

 **MATCH → SPOT → CHECK → WIN**

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